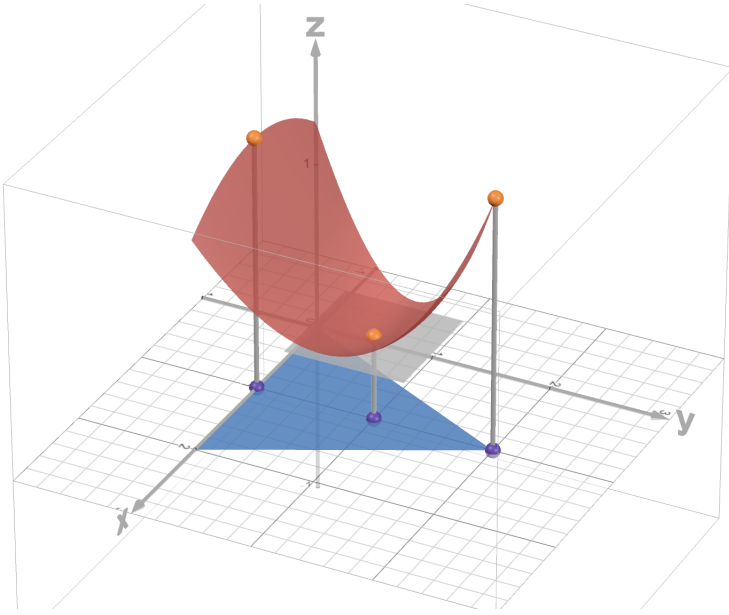


A. **Extremums.** Restrict inputs of differentiable $f(x, y)$ to **feasible set** $\mathcal{S}$.



We search for **extremums** of $f(x, y)$, aka:

In the sketch above, the **global** or **absolute maximum** occurs at inputs both

-
-

Inputs that yield maximums are called:

In the sketch above, the **global** or **absolute minimum** occurs at an input

-

Inputs that yield minimums are called:

Inputs where the tangent plane to $z = f(x, y)$ are horizontal are called

can also be identified by setting:

Extremizers of differentiable $f(x, y)$ on a feasible set $\mathcal{S}$ only occur

- either
- or

Example 1. Find the absolute maximum and minimum of f on $\mathcal{S}$.

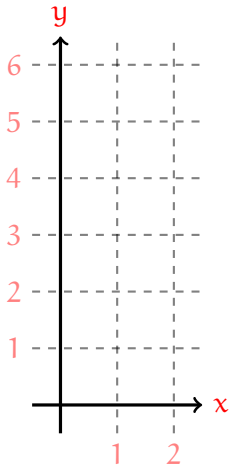
$$f(x, y) = x^2 - xy + y^2 - 3y$$

$$\mathcal{S} = \{(x, y) \mid 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 3x\}$$

Solution. At the moment we will assume

but later in the handout we will establish why they do.

First let us sketch the feasible set.



We have to examine the two types of inputs where extremizers could occur.

First we find critical points by setting

We observe that this point is:

and thus we box value

Next we address edges.

The bottom edge is defined by

We use this to write f as a one-variable function along this edge.

By inspection, the maximum and minimum on the bottom edge are

We continue with Example 1.

The diagonal edge is defined by

We use this to write f as a one-variable function along this edge.

To look for extremizers of this one-variable function we find critical points by

In our case

We observe that this point is in interval:

We compare the value of f here with the endpoints of the interval.

The vertical edge is defined by

We use this to write f as a one-variable function along this edge.

To look for extremizers of this one-variable function we find critical points by

In our case

We observe that this point is in interval:

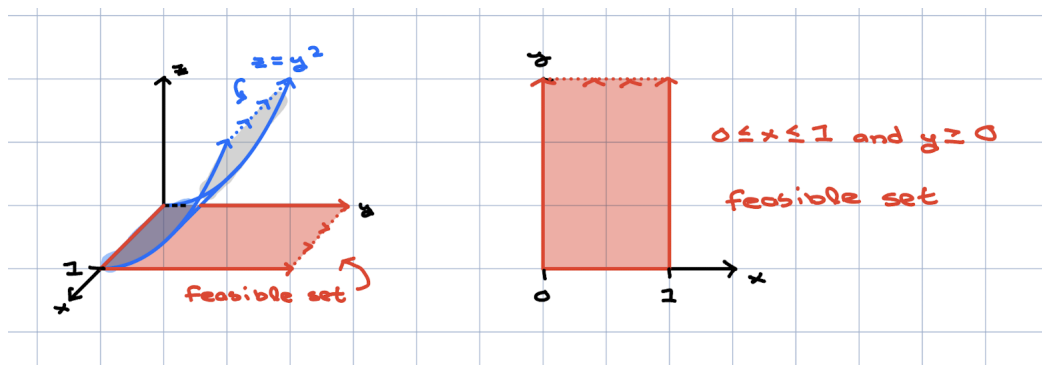
We compare the value of f here with the endpoints of the interval.

We have now obtained a complete list of candidates for extremums.

Comparing, we see the absolute extremums of f on $\mathcal{S}$ are

B. Extreme Value Theorem. When do extremums exist?

Consider function $z = y^2$ and feasible set $\mathcal{S} = \{(x, y) \mid 0 \leq x \leq 1, y \geq 0\}$ below.



Because y in $\mathcal{S}$ can be arbitrarily large so too can:

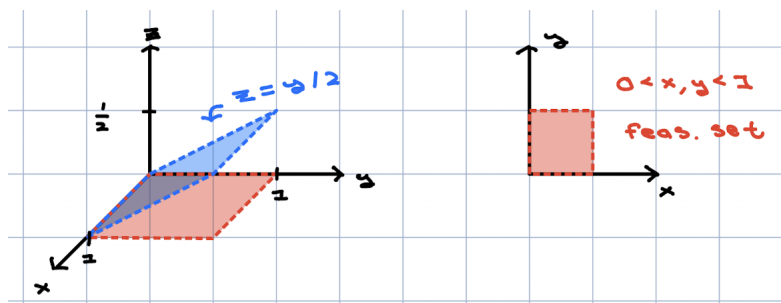
Therefore $z = y^2$ has no:

The issue is tied to the nature of the feasible set $\mathcal{S}$.

The feasible set $\mathcal{S}$ contains points that are

and thus the feasible set $\mathcal{S}$ is said to be:

Consider function $z = y/2$ and feasible set $\mathcal{S} = \{(x, y) \mid 0 < x, y < 1\}$ below.



Because y in $\mathcal{S}$ can be arbitrarily close, but not equal, to 1, function $z = y/2$ can

Therefore $z = y/2$ has no:

The issue is tied to the nature of the feasible set $\mathcal{S}$.

There are points on the **boundary** of the feasible set $\mathcal{S}$ that

and thus the feasible set $\mathcal{S}$ is said to be:

A set that is both **bounded** and **closed** is called:

Extreme Value Theorem. Let f be continuous and feasible set $\mathcal{S}$ compact. Then